

Syllabus

BTS 510 Syllabus – Fall 2026

Instructor information

- Stefany Coxe, Ph.D.
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 - Office hours: By appointment (send me an email to schedule)
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 - Office hours: TBD

Course information

Learning goals

This course covers *linear models* (i.e., regression) with an emphasis on **experimental studies**. Topics include using and manipulating datasets, plotting data, estimation and uncertainty, linear regression, logistic regression, Poisson regression, and survival analysis.

Learning objectives

- **Select** the appropriate linear model for the research question
- **Analyze** data with the appropriate linear model
- **Describe** your findings for scientific and non-scientific audiences
- **Create** a written report of your findings
- **Make conclusions** about your research question(s) based on those results

Course structure

Time and location

We will meet in-person on **Thursday** from *10am to 11:30am* in **BTS classroom on the 7th floor of PDC**.

Course format

This course takes place in a **flipped, hybrid format**. We will meet in person for 1 hour 30 minutes each week. This time will be spent on **hands-on statistical programming in R**. You will be responsible for completing course assignments such as videos and readings **prior to class** in order to be prepared to participate in the in-person meeting. You will also have assignments to complete after class.

Each week will follow a similar structure:

- Monday: Lecture videos posted
- Wednesday: Watch lecture and respond to survey by 8pm Wednesday
- Thursday: In-person meeting to review material and work on applications
- Sunday: Homework assignment due by end of day (midnight)

Software

We will be using **R** for this course. It's ok if you don't know how to use R (but also great if you do!). We'll start from scratch in the first few weeks. I will provide information about the specific procedures you will need to know for this course.

- R is free and open source and works on any platform (Windows, Mac, Unix)
 - Download R [here](#)
 - I also recommend using Rstudio, which can be downloaded [here](#)
 - If you use a Chromebook or tablet, you can use [Rstudio via the cloud](#)
 - * I have not used this much, so I don't know all its shortcomings
- We'll also be using [Quarto](#)
 - Publishing system based on markdown
 - Intersperse plain text and code
 - Output to convenient formats, like HTML, PDF, Word
 - [Install Quarto](#)

Assessments

Your work in this course will be assessed using a variety of methods.

Lecture videos (10%)

Watch the lecture video. Respond to the survey afterward with questions and comments.

Homework (60%)

There will be **five (5)** homework assignments. The assignments generally involve running analyses in R, interpreting output, and presenting the results.

Final project (30%)

You will select a dataset, propose some research questions that can be answered using linear models, write up the results, and present your findings to the class. More details to follow throughout the semester.

Tentative schedule

Week	Date	Topic	L	H
1	Sept 03	Introduction, R, markdown	1	
2	Sept 10	Data manipulation	2	1
3	Sept 17	Samples, sampling distribution	3	
4	Sept 24	ggplot	4	Data
5	Oct 01	ggplot	5	2
6	Oct 08	Linear regression	6	
7	Oct 15	Linear regression	7	3
8	Oct 22	Linear regression	8	
9	Oct 29	Linear regression	9	Proposal
10	Nov 05	Linear regression	10	4
11	Nov 12	Logistic regression	11	
12	Nov 19	Poisson regression	12	5
13	Nov 26	NO CLASS		
14	Dec 03	Survival analysis	13	
15	Dec 10	Repeated measures	14	
16	Dec 17	Presentations		Final

- **L:** Watch lecture videos and respond to survey by **Wednesday at 8pm.**

- **H:** Homework assignments are due by the **end of Sunday (midnight)**.
- **Final project**
 - **Data:** Description of data due by September 27
 - **Proposal:** Proposal of final project due by November 1
 - **Presentation:** Presentation in class on December 17
 - **Paper:** Final paper due by December 20
- **All due dates subject to change** due to weather, emergency, scheduling changes, etc.

Grades

Grade	Percentage
A+	≥ 97
A	93 - 96.99
A-	90 - 92.99
B+	87 - 89.99
B	83 - 86.99
B-	80 - 82.99
C+	77 - 79.99
C	70 - 76.99
D+	65 - 69.99
D	60 - 64.99
F	≤ 59.99

Cedars-Sinai policies

Attendance

Attendance is not explicitly part of your grade in this course, but activities completed during the in-person portion of the course will be **very** helpful. If you need to miss class (such as for illness, religious event, professional activity, university-sanctioned event, or **any other reason**), please contact me to make any necessary arrangements.

Academic dishonesty and misconduct

Please refer to your [policy handbook](#) for a description of what constitutes academic dishonesty.

- While you may work with other students on your homework assignments, I expect all students to complete and turn in their own work.